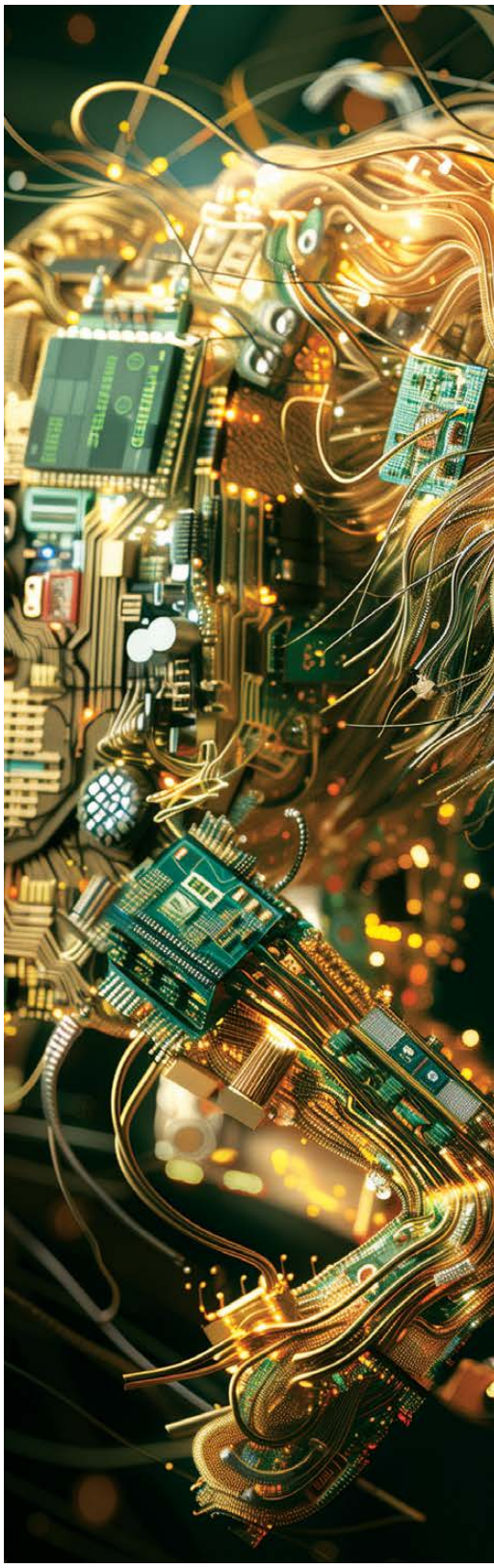


By **Ashok Jhunjhunwala**

STRIVE TO BE TECHNOLOGY LEADERS

To propel India's ascent, we should harness its strengths in human capital and tackle pressing societal problems through tech-based solutions



The 'Make in India' programme has given a push to some local manufacturing, but it is merely an assembly job with imported technology and components. The result is a low value-addition. As we aim to become a developed nation, we need to carry out design and development in India, use as many local components as possible, make machinery for production and own a significant part of IPR (intellectual property right) for the products. We need to strive to be amongst the technology leaders in at least a few areas. How do we go about doing it? We need to first understand India's strengths:

- We now have an increasing number of young scientists and engineers with high-quality expertise, having obtained the best training available. They are raring to go and to lead, if the nation can provide the right push and support.
- We have a large pool of fresh engineering graduates, and they can be good even if they come from lesser known colleges. They must be motivated to work hard for long hours, learn from each other, and take up impossible challenges. Fortunately, the cost of such talent in India is much lower than elsewhere.
- We have a very large market, but only for affordable products. This makes the task of developing such products a challenge, but overcoming it would push us towards a leadership position in technology.

Given these strengths, what do we do? We need to, of course, invest in R&D and design products. But when we invest say $100 million in AI, countries like the US would invest several hundred billion dollars. Trying to compete head-on with the West in areas with very large capital investments is not going to get us far. Instead, we need to focus on some big problems that our society faces and for which we have no solution at present. If we think out of the box and bring together technology expertise from multiple domains, we will create momentum in the right direction. We would have to work extremely hard, be ready to fail, fall and get up again to be able to make progress. As we solve such impossible problems, we will emerge as technology leaders in multiple areas. Let me illustrate this with two examples.

Illustration by **NILANJAN DAS**